

JavaScript Basics

1. Write a program which consist of single line and multiline comments.

Answer:

```
<!DOCTYPE html>
<html>
<head>
    <title>JavaScript Comments</title>
</head>

<body>

<h2>JavaScript Comments Example</h2>

<script>

// This is a single-line comment

/*
This is a
multi-line comment
*/

document.write("JavaScript Comments Example");

</script>

</body>
</html>
```

2. Create an array consisting of fruits names and when we click on a button sort it in descending order.

Answer:

```
<!DOCTYPE html>
<html>
<head>
    <title>Array Sorting</title>
</head>

<body>

<h2>Sort Fruits in Descending Order</h2>

<button onclick="sortFruits()">Sort Fruits</button>

<p id="result"></p>

<script>

let fruits = ["Apple", "Orange", "Banana", "Mango", "Grapes"];

function sortFruits() {
    fruits.sort();
    fruits.reverse();
}
```

```
        document.getElementById("result").innerHTML = fruits.join(", ");
    }

</script>

</body>
</html>
```

3. Write a program with FOR/IN loop.

Answer:

```
<!DOCTYPE html>
<html>
<head>
    <title>For/In Loop</title>
</head>

<body>

<script>

let student = {
    name: "Suman",
    age: 21,
    course: "BCA"
};

for(let key in student){
    document.write(key + " : " + student[key] + "<br>");
}

</script>

</body>
</html>
```

4. Create an object "person" with properties firstname and lastname. Display these properties using 2 different ways of accessing.

Answer:

```
<!DOCTYPE html>
<html>
<head>
    <title>Object Example</title>
</head>

<body>

<script>

let person = {
    firstname: "Suman",
    lastname: "Kumar"
};

// Dot notation
```

```

document.write(person.firstname + "<br>");

// Bracket notation
document.write(person["lastname"]);

</script>

</body>
</html>

```

5. Write a program with variable hoisting(initialization first then declaration).

Answer:

```

<!DOCTYPE html>
<html>
<head>
    <title>Variable Hoisting</title>
</head>

<body>

<script>

x = 100;        // Initialization before declaration
document.write(x + "<br>");

var x;          // Declaration

</script>

</body>
</html>

```

Explanation:

Variables declared with var are **hoisted**, meaning the declaration is moved to the top of its scope during execution. Only the declaration is hoisted, not the initialization.

6. Use strict mode in your program and understand why variable not declared causes error.

Answer:

```

<!DOCTYPE html>
<html>
<head>
    <title>Strict Mode</title>
</head>

<body>

<script>

"use strict";

x = 10;    // Error: x is not declared

document.write(x);

```

```
</script>
```

```
</body>
```

```
</html>
```

Explanation :

- `"use strict";` enables **strict mode**.
- In strict mode, variables **must be declared** using `let`, `const`, or `var`.
- Writing `x = 10;` without declaring `x` causes a **ReferenceError**.

Correct Version:

```
"use strict";  
let x = 10;  
document.write(x);
```

Things to know:

1. How JavaScript differs from Java?

Answer:

JavaScript	Java
Scripting language	Programming language
Interpreted by the browser (or JavaScript engine)	Compiled into bytecode and runs on the Java Virtual Machine (JVM)
Primarily used for web development and adding interactivity	Used for desktop, web, mobile, and enterprise applications
Dynamically typed (variable types are determined at runtime)	Statically typed (variable types must be declared)
Prototype-based object model	Class-based object-oriented programming
Runs directly in web browsers	Requires the Java Runtime Environment (JRE) or JVM
Easy to integrate with HTML and CSS	Independent of HTML and CSS
File extension: .js	File extension: .java